Paul-Henry Leemann

Assistant professor in pure mathematics

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Personal Information

Date of Birth April 12, 1987
Citizenship Switzerland

Main areas of research

My research interests lie at the intersection of geometric and combinatorial group theory, graph theory and symbolic dynamic. I am particularly interested in groups acting on rooted trees and their (weakly) maximal subgroups, limits and coverings of (Schreier) graphs, rigidity phenomenon in Cayley and Schreier graphs and graphs associated to dynamical systems.

Employment

Since 2022 **Xi'an Jiaotong-Liverpool University, Suzhou, China**, *Assistant Professor*, in Department of Pure Mathematics

2021–2022 **Université de Neuchâtel, Switzerland**, *Scientific collaborator*, in Institut de mathématiques

2020–2021 **Université de Neuchâtel, Switzerland**, *Post-Doc*, in Institut de mathématiques, with Alain Valette

2018–2020 **ENS-Lyon, France**, *Post-Doc*, in Unité de Mathématiques Pures et Appliquées, with Mikael de la Salle
Post-Doc Labex MILYON

2016–2018 **ENS-Lyon, France**, *Post-Doc*, in Unité de Mathématiques Pures et Appliquées, with Mikael de la Salle
Early Postdoc.Mobility grant supported by the Swiss National Science Foundation

Education

2011–2016 **Université de Genève, Switzerland**, *Ph.D. in Mathematics*, "On Subgroups and Schreier Graphs of Finitely Generated Groups", advisor: Tatiana Smirnova-Nagnibeda
Mention very good. **Vacheron Constantin prize for the best thesis in sciences, of the Université de Genève**

2013–2015 **Université de Genève, Switzerland**, *Basic Certificate of Didactic in Mathematics and in Educational Science*

2009–2011 **Université de Genève, Switzerland**, *M. Sc. in Mathematics*, Master-thesis: "Caractères irréductibles des groupes de Coxeter finis", advisors: Pierre de la Harpe and Tatiana Smirnova-Nagnibeda
grades: 6/6 (Master-thesis) and 5.7/6 (courses)

2006–2009 **Université de Genève, Switzerland**, *B. Sc. in Mathematics*

Published papers

15. *Complexity functions of infinite sequences*, with B. Greenfeld, C. Gustavo Moreira and E. Zelmanov. Proc. Natl. Acad. Sci. U.S.A. Vol. 123 No. 44 (2026). https://doi.org/10.1073/pnas.2627766123
14. *Limits of Rauzy graphs of languages with subexponential complexity*, with T. Nagnibeda, A. Skripchenko and G. Veprev. Groups, Geometry, and Dynamics (2026), published online first. https://doi.org/10.4171/GGD/1009
13. *On the structure of finitely generated subgroups of branch groups*, with D. Francoeur, R. Grigorchuk and T. Nagnibeda. Journal of Combinatorial Algebra (2025), published online first. https://doi.org/10.4171/JCA/112
12. *Weakly maximal subgroups of branch groups.* International Journal of Algebra and Computation, Vol. 35 Issue 05 (2025). https://doi.org/10.1142/S0218196725500158
11. *Subgroup induction property for branch groups*, with D. Francoeur. Journal of Fractal Geometry, Vol. 12 Issue 1 (2025). https://doi.org/10.4171/JFG/162
10. *Wreath products of groups acting with bounded orbits*, with G. Schneeberger. L'Enseignement mathématiques, Vol. 70 Issue 1 (2024). https://doi.org/10.4171/LEM/1059
9. *Most rigid representation and Cayley index of finitely generated groups*, With M. de la Salle. The Electronic Journal of Combinatorics, Vol. 29 Issue 4 (2022). https://doi.org/10.37236/10512
8. *Property FW and wreath products of groups: a simple approach using Schreier graphs*, with G. Schneeberger. Expositiones Mathematicae, Vol. 40 Issue 4 (2022). https://doi.org/10.1016/j.exmath.2022.07.001
7. *Cayley graphs with few automorphisms: the case of infinite groups*, with M. de la Salle. Annales Henri Lebesgue Vol. 5 (2022). https://doi.org/10.5802/ahl.118
6. *Up to a double cover, every regular connected graph is isomorphic to a Schreier graph.* Bulletin of the Belgian Mathematical Society – Simon Stevin Vol. 28, Issue 3 (2022). https://doi.org/10.36045/j.bbms.210416
5. *Finitely generated subgroups of branch groups and subdirect products of just infinite groups*, with R. Grigorchuk and T. Nagnibeda. Izvestia: Mathematics Vol. 85 (2021). https://doi.org/10.1070/IM9101
4. *Cayley graphs with few automorphisms*, with M. de la Salle. Journal of Algebraic Combinatorics Vol. 53 (2021). https://doi.org/10.1007/s10801-020-00956-1
3. *Schreier graphs: Transitivity and Coverings.* International Journal of Algebra and Computation Vol. 26, Issue 1 (2016). https://doi.org/10.1142/S021819671650003X
2. *Weakly maximal subgroups in regular branch groups*, with K. Bou-Rabee and T. Nagnibeda. Journal of Algebra Vol. 455 (2016). https://doi.org/10.1016/j.jalgebra.2016.02.009
1. *Lamplighter groups, de Bruijn graphs, spider-web graphs and their spectra*, with R. Grigorchuk and T. Nagnibeda. Journal of Physics A: Mathematical and Theoretical Vol. 49, Number 20 (2016). https://doi.org/10.1088/1751-8113/49/20/205004

Preprints

17. *Polynomial complexity, random graphs, and strict ergodicity*, with B. Greenfeld, C. Gustavo Moreira and E. Zelmanov. (2026) https://arxiv.org/abs/2611.08973
16. *Non-closed subgroups of weakly branch groups*, with J. Fariña-Asategui and T. Nagnibeda. (2025). https://arxiv.org/abs/2511.15277

Prizes and Grants

2025–2026 *Constructing a comprehensive and cohesive advanced linear algebra textbook tailored to XJTLU's curriculum.* School of mathematics and physics Teaching Grant, XJTLU (China).

2024–2026 *Groups acting on rooted trees and their subgroups.* XJTLU Research Development Fund (China).

2023–2027 *Subgroup separability and rigidity of Cayley graphs.* Bourse Beatriz Galindo (junior). 4 years research position at the Universidad del País Vasco (Bilbao, Spain). Declined.

2021–2022 *Group theory, topology and dynamics: Russian-Swiss collaboration.* Co-investigator, 2 years collaboration

2018–2020 *Local and global geometry of graphs, groups and their actions.* Labex MILYON Postdoc grant

2016–2018 *Local and global geometry of graphs, groups and their actions.* Early Postdoc.Mobility grant supported by the Swiss National Science Foundation

2017 Vacheron Constantin prize for the best thesis in sciences, *Université de Genève, Switzerland*

Conferences organisation

June 2022 Self-similarity of groups, trees and fractals, co-organiser, 30 May-3 June, Institut Henri Poincaré, Paris, France

October 2021 Kervaire seminar "Baum-Connes conjecture", co-organiser, 24-29 Octobre, Les Diablerets, Switzerland

March 2013 Kervaire seminar "Géométrie des groupes 2013", local organizer, 10-15 March, Les Diablerets, Switzerland

Talks

Conferences:

July 2025 The 15th Ukraine Algebra Conference, National University of Lviv, Lviv, Ukraine, online presentation, *On the structure of finitely generated subgroups of branch groups.*

August 2021 World Of Group Craft, online conference, *Wreath product of groups acting with bounded orbits.*

June 2018 Trees Dynamics and Locally Compact Groups, Heinrich Heine University, Dusseldörf, Germany. *Profinite completion of weakly maximal subgroups of branch groups.*

January 2017 Rencontre GAMME, Sète, France. *Weakly maximal subgroups of the Grigorchuk group. Existence and construction of IRS.*

November 2014 Paroles aux jeunes chercheurs en géométrie et dynamique, GDR Platon, Université de Bordeaux, France. *A criterion for the transitivity of Schreier graphs.*

June 2014 Workshop in Algebra and Geometry, University of Bern, Switzerland. *Transitivity of Schreier graphs and strongly simple groups.*

September 2013 Geometric and Analytic Group Theory, Ventotene, Italy. *Transitivity of Schreier graphs.*

Seminars (since 2019):

June 2026 Probability Seminar, National Taiwan University, Taipei, Taiwan. *Probabilistic limit of Rauzy graphs*

February 2026 Séminaire Groupes et Géométrie, Université de Genève, Switzerland. *Cayley graphs with few automorphisms: beyond finitely generated groups*

December 2025 International Center for Mathematics Seminar, SuSTech, China. *Limits of Rauzy graphs*

May 2025 Geometric Group Theory Seminar, Harbin Institute of Technology, China. *Maximal subgroups, weakly maximal subgroups and generalisations.*

February 2025 Séminaire Groupes et Géométrie, Université de Genève, Switzerland. *Sous-groupes maximaux, faiblement maximaux et généralisations.*

June 2024 Geometry, algebra and operator algebra seminar, Université Clermont Auvergne, Clermont-Ferrand, France. *Nouveaux examples de groupes séparables par les sous-groupes (LERF).*

April 2024 Topology seminar, XJTLU, Suzhou, China. *Branch groups and the LERF property.*

April 2024 Topology seminar, XJTLU, Suzhou, China. *Around the profinite topology on groups.*

November 2023 Topology seminar, University of Fudan, Shanghai, China. *Rigidity of Cayley graphs.*

October 2023 Mathematics Colloquium, National Taiwan University, Taipei, Taiwan. *Random walks on infinite groups: some applications.*

September 2022 Pure Math seminar, XJTLU, Suzhou, China. *Cayley graphs with few automorphisms.*

March 2022 Colloquium of the Université de Fribourg, Switzerland. *Cayley graphs with few automorphisms.*

June 2021 Journées Neuchâtel-Besançon d'Analyse Fonctionnelle, Université de Neuchâtel, Switzerland. *Produit en couronne et groupes agissant avec des orbites bornées.*

May 2021 Séminaire Groupes et Géométrie, Université de Genève, Switzerland. *Produit en couronne et groupes agissant avec des orbites bornées.*

April 2021 Algebra Seminar, Lincoln University, United-Kingdom. *Branch groups and the subgroup induction property.* Slides: https://www.leemann.website/slides/subgroupinduction.pdf

April 2021 Séminaire Géométrie, Algèbre, Algèbre d'opérateurs, Université de Clermont-Ferrand, France. *Graphes de Cayley avec peu d'automorphismes.* Slides (in French): https://www.leemann.website/slides/clermont-ferrand.pdf

March 2021 Group theory seminar, ENS, Paris, France. *De Bruijn graphs, spider web graphs and Lamplighter groups.* Slides: https://www.leemann.website/slides/bruijn.pdf

February 2021 Symmetry in Newcastle, University of Newcastle, Australia. *Rigidity of Cayley graphs.* Slides: https://www.leemann.website/slides/newcastle.pdf

December 2020 Séminaire Groupes et Analyse, Université de Neuchâtel, Switzerland. *Property FW and wreath products, the median graphs point of view*

November 2020 Colloque du département de mathématiques, Université de Neuchâtel, Switzerland. *Groups and graphs: rigidity phenomena.* Slides (in French): https://www.leemann.website/slides/rigidity.pdf

March 2020 Séminaire Groupes et Géométrie, Université de Genève, Switzerland. *Weakly maximal subgroups of branch groups.* Slides: https://www.leemann.website/slides/weaklymaximal.pdf

October 2019 Séminaire de Géométrie, Groupes et Dynamiques, ENS Lyon. *Rigidity of Cayley graphs.*

July 2019 Séminaire Darboux, Montpellier, France. *Rigidity of Cayley graphs.*

Teaching Experiences

Since 2022	Lecturer at XJTLU:
1x	Metric spaces: metric spaces, continuity, connectedness, compactness, completeness (2nd year class). Lecture notes https://www.leemann.website/cours/MTH224.pdf and tutorials https://www.leemann.website/cours/MTH224-Tutorials.pdf
2x	Advanced Linear Algebra: vector spaces, linear maps, inner product spaces, eigen theory (1st year class). Lecture notes https://www.leemann.website/cours/MTH107.pdf and tutorials https://www.leemann.website/cours/MTH107-Tutorials.pdf
1x	Dynamic Modelling: discrete and continuous-time population models (1st year class)
2x	Engineering Mathematics: complex analysis, Laplace transform, ODEs (1st year class)
2x	Introduction to the Methods of Applied Mathematics: ODEs, Fourier series, PDEs (1st year class)
13x	Supervisor of Final Year Projects (Bachelor)
2020–2021	In Neuchâtel:
1x	"Branch groups", a 6 hours mini-course for doctoral students
2016–2020	In ENS Lyon:
3x	Teaching assistant of "Coverings and fundamental groups" (master)
1x	Teacher of "Introduction to LaTeX" (3rd year class)
2011–2016	Teaching assistant at the University of Genève:
1x	Advanced course on Combinatorics (master)
3x	Algebra II: groups, rings, fields, Galois theory (2nd year class)
2x	Algebra I: linear algebra (1st year class)
1x	Analysis I: basics of logic and set theory, real analysis (1st year class)
1x	General mathematics: 1st year class for students of other departments
1x	Tutoring for first year students
2009–2010	Graduate teaching assistant at the University of Genève: tutoring for first year students
2006–2011	Private tutor in mathematics for high school students

Scientific popularization

2018	MATHαLYON: interactive workshop in Junior High School, Lyon, France
2015	Facilitator at the "Science's night", Genève, Switzerland
2012–2014	Facilitator at the University "Open Day", Genève, Switzerland
2011–2012	Mathematics' facilitator in Junior High School and High School classes, Genève, Switzerland
2010	Facilitator at the *Castle of games*, Tour-de-Peilz (Suisse),
2009–2010	Mentor in the "boussole"curriculum (one week of immersion in the mathematics' department, for High School's students), Genève, Switzerland

Administrative duties

Since 2024	Department Exam Officer, Department of Pure Mathematics, XJTLU, China
Since 2024	Member of the School Learning and Teaching Committee, School of Mathematics and Physic, XJTLU, China
2022–2023	Global Engagement Officer, Department of Pure Mathematics, XJTLU, China

2022–2023 Member of the School International Committee, School of Mathematics and Physic, XJTLU, China

2022–2023 Member of the School International Strategy Working Group Committee, School of Mathematics and Physic, XJTLU, China

Memberships in panels and scientific reviewing activities

I have been an external expert for the OPUS-21 grant funding scheme of the National Science Centre Poland.
I am a reviewer for Mathematical Reviews/MathSciNet.
I have been a referee for the following journals (classified by date of first review): L'enseignement mathématiques, Groups, Geometry, and Dynamics, Journal of Algebra and its Applications, Journal of Algebra, International Journal of Algebra and Computation, Expositiones Mathematicae, The Australasian Journal of Combinatorics, Panorama et Synthèses (SMF), Experimental Mathematics, The Art of Discrete and Applied Mathematics, Journal of Computational Algebra, International Mathematics Research Notices.

Languages

French: native English: fluent German: basics Mandarin: beginner

Programming skills

(LA)TEX and Ti*k*Z: advanced Lua: basics HTML/CSS: basics GAP: basics

(The background pictures are computed in Lua and rendered in Ti*k*Z.)